DISASTER HEALTHCARE AI - EXECUTIVE SITUATION REPORT

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1. Executive Summary & Incident Status

This report presents real-time AI-driven healthcare demand predictions and multi-objective resource allocation plans for Maharashtra's 35 administrative districts. Under current disaster parameters, 2 districts are classified at HIGH SHORTAGE RISK. The OR-Tools SCIP optimization solver recommends transferring 1,000 units of essential medical supplies across 1 transit routes.

2. High Shortage-Risk District Priorities

District	Medicine	Demand (P50)	Surge Stock	P(Shortage)
Raigad	blood_units	2,112	2,200	47.5% (HIGH)
Satara	blood_units	2,135	2,528	40.8% (HIGH)

3. Recommended Inter-District Resource Transfers

Source District	Dest District	Medicine	Quantity	Distance (km)
Pune	Sangli	antibiotics	1,000	210.0 km

4. Data Provenance & Method Disclosures

Every figure in this report carries strict provenance verification:

- [REAL]: EM-DAT (CRED), National Hospital Directory (data.gov.in), WorldPop 2020 Grid.
- [ESTIMATED]: District bed capacity (scaled from 11,587 state ICU bed audit), event weather features.
- [ASSUMED]: Epidemiological daily demand rates from WHO IEHK 2017 & Sphere Handbook standards.

Model outputs are non-binding recommendations requiring human officer validation.